

Break Time Tracker

Complete User Guide

QR Code-based Break Time Management System
For Operators, Supervisors, Team Leaders, Coordinators, and Admins

OPERATOR

Requests breaks
via QR scan

SUPERVISOR / TL

Reviews & approves
break requests

ADMIN

Manages users,
settings & reports

1. Introduction

The Break Time Tracker is a full-stack web application designed to streamline break time management for teams and organizations. It uses QR code scanning to allow operators to request breaks quickly, while supervisors and team leaders review and approve requests with custom durations. The system provides real-time dashboards, automated reminders, and comprehensive reports.

This guide covers everything you need to know to use the application effectively, from your first login to generating daily reports.

2. User Roles & Permissions

The application uses role-based access control. Every user logs in through the same login page, but the features available depend on their assigned role.

Role	Primary Responsibilities	Key Access
Operator	Request breaks via QR scan; return on time	Scan, Dashboard, History
Supervisor / TL / Coordinator	Review, approve, or reject break requests	Dashboard (approve/reject), Reports
Admin	Manage users, devices, settings; view all logs	Users, Devices, Settings, Reports

Important: All roles share the same login page. Your username and password determine your role and available features automatically.

3. Break Request Flow

The following diagram illustrates the complete lifecycle of a break request, from login to completion.

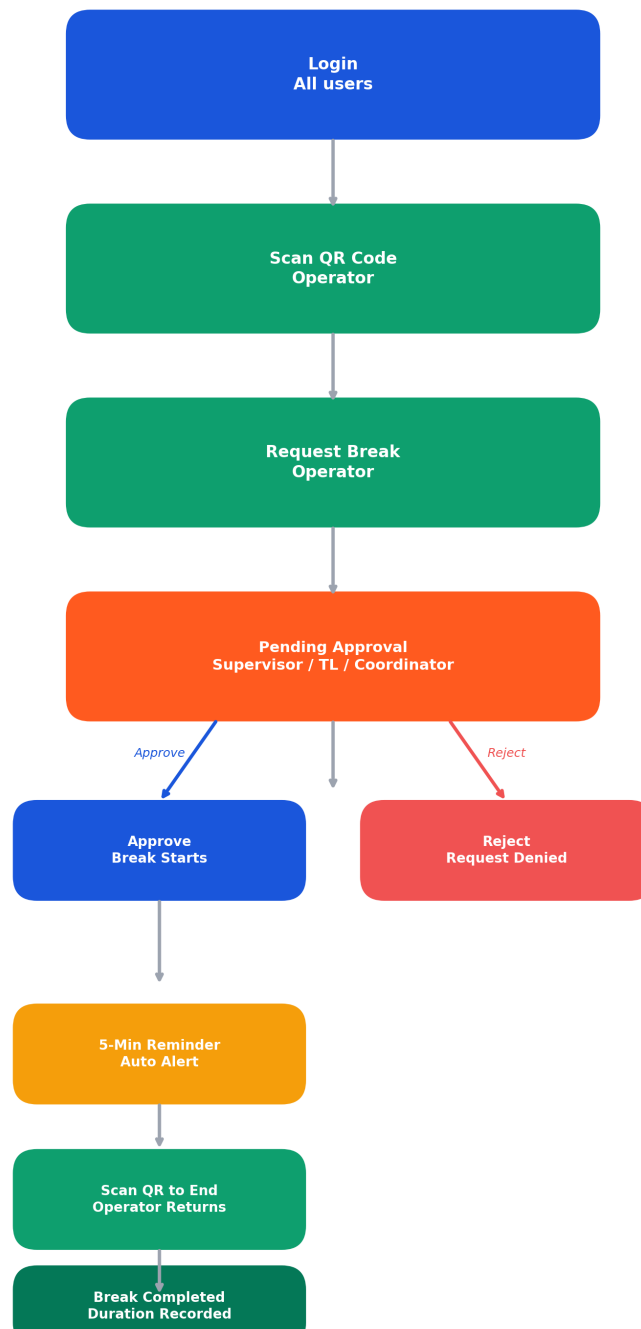


Figure 1: Break Request Flow Diagram

4. Step-by-Step User Guide

4.1 Logging In

Open the application in your web browser. You will see a single login screen for all users. Enter your username and password, then click the Login button.

Default Admin Credentials:

```
Username: admin
Password: Admin123
```

The admin account is automatically created when the server starts for the first time. Use this account to create additional users for your team.

4.2 Requesting a Break (Operator)

Operators can request breaks using the QR code scanner or directly from the Dashboard.

- Navigate to the **Scan** page from the sidebar menu.
- Tap the **Request Break** button.
- Point your device's camera at the QR code placed at the break area.
- The request is sent to your Supervisor / Team Leader / Coordinator for approval.
- Wait for approval. You will see your request status update in real time.

Alternative: You can also request a break directly from the Dashboard without scanning a QR code. The QR code is simply the fastest method.

4.3 Approving or Rejecting Requests (Supervisor / TL / Coordinator)

When an operator requests a break, it appears on the Dashboard as a pending request.

- Open the **Dashboard** page.
- Locate the pending break request in the 'Pending Requests' section.
- Review the operator's name and current status.
- Set a custom break duration (in minutes) using the duration field.
- Click **Approve** to allow the break, or **Reject** to deny it.
- The operator is notified immediately of your decision.

4.4 During the Break

Once a break is approved, the system automatically starts a countdown timer. The operator's status changes to 'On Break' and is visible on the live Dashboard to all team members.

5-Minute Reminder: The app sends an automatic notification to the operator 5 minutes before their approved break duration ends. This gives them enough time to wrap up and return to their station.

4.5 Ending a Break (Operator)

When the operator returns from their break, they must mark it as ended.

- Return to the break area QR code.
- Navigate to the **Scan** page and tap **End Break**.
- Scan the QR code to confirm your return.
- The system records the exact return time and marks the break as 'Completed'.

Note: You can also end the break from the Dashboard if you are unable to scan the QR code.

5. App Screens Overview

The application is organized into several screens, each serving a specific purpose. Below is a summary of every screen and who can access it.

Screen	Description	Who Can Access
Dashboard	Live break status, pending requests, staffing coverage	Everyone
Scan	QR code scanner for quick break request and end	Everyone
History	Personal break history for the current day	Everyone
Reports	Daily break statistics, late returns, staff summaries	Everyone
Users	Add, edit, and remove team members	Admin, Supervisor
Devices	Manage registered devices and access control	Admin, Supervisor

6. Key Features Explained

6.1 QR Code Scanning

Place a printed QR code at each break area. Operators simply point their device camera at the code to request or end breaks. No typing or manual entry is required, reducing errors and saving time. The system validates the scan and links it to the operator's account.

6.2 Approval Workflow

Every break request goes through an approval process. Supervisors, Team Leaders, and Coordinators can set custom break durations per request, ensuring staffing coverage is maintained. Approved breaks start automatically; rejected requests notify the operator immediately.

6.3 Real-Time Dashboard

The Dashboard provides a live view of all break activity: who is on break, who has pending requests, and overall staffing coverage. It updates in real time without requiring a page refresh, so supervisors always have current information.

6.4 Smart Reminders

Operators receive an automatic notification 5 minutes before their break ends. This helps prevent late returns and keeps operations running smoothly. Late returns are flagged in reports for follow-up.

6.5 Reports & Analytics

The Reports screen provides daily summaries including total break counts, average break duration, late return statistics, and per-staff breakdowns. Use these insights to identify patterns and optimize team scheduling.

6.6 Shift Management

The system supports multiple shift types: Morning, Afternoon, Night, and Rotating. Break policies and reports can be filtered by shift, making it easy to manage teams that work across different time periods.

6.7 Password Management

All users can change their own password from the Settings menu. Admins can also reset passwords for other users if needed. Passwords are stored securely using industry-standard hashing.

7. Tips & Best Practices

Keep QR Codes Visible: Ensure QR codes at break areas are printed clearly, protected from damage, and placed at a comfortable scanning height.

Approve Promptly: Supervisors should check the Dashboard regularly and approve requests quickly to minimize operator waiting time.

Set Realistic Durations: When approving breaks, set durations that match your team's operational needs while complying with labor regulations.

Monitor Late Returns: Use the Reports screen to track late returns. Address patterns early to maintain discipline and coverage.

Update Settings as Needed: Admins should review app settings periodically to ensure break policies and shift configurations remain accurate.

Register Devices: For added security, use the Devices screen to register and manage which devices can access the application.

Pro Tip: Operators can use the Dashboard as a backup if the QR code scanner is unavailable. Both methods achieve the same result.

8. Troubleshooting

Issue	Solution
Cannot scan QR code	Ensure camera permissions are enabled. Try the Dashboard button as a backup.
Break request not appearing	Check your internet connection. Refresh the Dashboard page.
Approval buttons not visible	Your account may have Operator role only. Contact an Admin to verify.
Forgot password	Contact your Admin to reset your password from the Users screen.
Page not loading	Clear browser cache or try a different browser. Check server status.

For technical support or feature requests, contact your system administrator.
Built with React, Node.js, Express, and MongoDB.